

Plot points in four quadrants

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Plot A(-4, 3). Which quadrant contains A? Copy or complete the first step.

2. Point P moves from (2, -5) to (-3, 1). Describe the translation.

3. Translate (4, -2) by 3 left and 5 up.

4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Plot A(-4, 3). Which quadrant contains A? Copy or complete the first step.
Answer Quadrant II.
2. Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer 5 left and 6 up.
3. Translate (4, -2) by 3 left and 5 up.
Answer (1, 3).
4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.

Plot points in four quadrants

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Plot A(-4, 3). Which quadrant contains A?

6. Check this result using an inverse, estimate or known fact:
Translate (4, -2) by 3 left and 5 up.

2. Point P moves from (2, -5) to (-3, 1). Describe the translation.

7. Prove or explain your reasoning: Reflect (-6, 2) in the y-axis.

3. Translate (4, -2) by 3 left and 5 up.

8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.

4. Solve this independently and show every stage: Plot A(-4, 3). Which quadrant contains A?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Point P moves from (2, -5) to (-3, 1). Describe the translation.

10. Write and solve a similar question to: Translate (4, -2) by 3 left and 5 up.

Teacher answers and checking prompts

1. Plot A(-4, 3). Which quadrant contains A?
Answer Quadrant II.
2. Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer 5 left and 6 up.
3. Translate (4, -2) by 3 left and 5 up.
Answer (1, 3).
4. Solve this independently and show every stage: Plot A(-4, 3). Which quadrant contains A?
Answer Quadrant II.
5. Use a second method or representation for: Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer Expected result: 5 left and 6 up. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Translate (4, -2) by 3 left and 5 up.
Answer Expected result: (1, 3). A valid check must be shown.
7. Prove or explain your reasoning: Reflect (-6, 2) in the y-axis.
Answer (6, 2). The explanation must show why the method works.
8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Translate (4, -2) by 3 left and 5 up.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Reflect $(-6, 2)$ in the y-axis.

6. Create a new example that tests the same idea as: Translate $(4, -2)$ by 3 left and 5 up.

2. Prove the answer to this question, rather than only calculating it: Plot $A(-4, 3)$. Which quadrant contains A?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Point P moves from $(2, -5)$ to $(-3, 1)$. Describe the translation.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $(1, 3)$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Reflect $(-6, 2)$ in the y-axis.
Answer $(6, 2)$. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Plot $A(-4, 3)$. Which quadrant contains A?
Answer Expected result: Quadrant II. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Point P moves from $(2, -5)$ to $(-3, 1)$. Describe the translation.
Answer Expected result: 5 left and 6 up. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $(1, 3)$.
Answer One valid reconstruction is: Translate $(4, -2)$ by 3 left and 5 up. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
6. Create a new example that tests the same idea as: Translate $(4, -2)$ by 3 left and 5 up.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.